

David Shontz

Professor West

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Natural Sciences Literature Review

In 1964, scientists discovered they could insert DNA into foods to genetically modify them, creating common practice in many products and foods to this day. Most of the genetically modified crops grown today were developed to help farmers prevent crop and food loss while also controlling weeds. The three most common traits found in genetically modified crops are: Resistance to certain damaging insects and Tolerance of certain herbicides used to control weeds. From the surface, this seems like a wonderful innovation, causing more food and better food to be grown. Unfortunately, the long-term side effects of genetically modified foods are still controversial, with some people explaining that they cause disease, and others claiming they have no long-term health effects. My sources will be organized thematically, based on the conclusions of the article.

Two specific articles claim that genetically modified foods come with serious dangers and side effects and warn the general public to not consume them. An article published to the National Center for Biotechnology Information website by A.S Bawa and K.R Anilakumar that explains the dangers of genetically modified foods. The article states that the worst effects genetically modified are on the human body. In some instances, causing diseases that are immune to antibiotics. Along with these dangers, people are every skeptical on the long-term

effects these genetically modified foods can have on the human body. The article continues to say that this is only the dangers at face value, and that there is still much research to be done to see the true length of dangers that genetically modified foods can cause. The writings of Bawa and Anilakumar are very similar to the writings of an article published by Cary Funk and Brian Kennedy to the Pew Research Website. Funk and Brian spent lots of time researching how regular foods differ from genetically modified foods. Along with this research they conducted surveys using interview style questions answered by the general public on their opinions of genetically modified foods. "When asked which of three positions best fits their viewpoints, about half of Americans (48%) say the health effects of GM foods are no different than other foods, 39% say GM foods are worse for one's health and one-in-ten (10%) say such foods are better for one's health." The article goes on to explain that the people who are aware of what GMO's are and how they work, care a significant amount more about them, and the risks that come with them. The article states, "People who have heard or read more about GM foods are much more likely to consider these foods worse for one's health." This supports the findings in the first article, explaining how genetically modified foods are bad for a person's health by showing that the people who have done their research agree with the negative correlation.

The other two articles claim that genetically modified foods have no adverse health effects, including long-term. An article published by Stefania Manetti to the National Library of Medicine website, explains the false information about the health risks in genetically modified foods. Manetti explains that genetically modifying a plant involves taking the gene of a desired plant or animal and inserting that gene into the DNA of another plant or animal. Genes can also be moved from animal to plant and vice versa. Manetti claims that there are no side effects to

consuming GMO's but gives a list of the benefits which include: more nutritious food, tastier food, increased food supply with a longer shelf life, and less use of pesticides. Manetti heavily believes that the research conducted is proven to show no side effects of genetically modified foods. Manetti's research and perspective is supported by the opinion of Jason Doug. Doug gave a response on a Quora forum on the subject. Jason, having done tons of research himself is trying to give a public opinions perspective and clear up misinformation, in the same way Manetti attempted to. Doug believes that the problem stems from American media and politicians commonly spreading misinformation on the matter. He claims that the media plays at people's fears, warning about "monstrous vegetables", or "poisoned fruit", when in reality, study after study has shown that GMO's have no harmful differences from non-GMO's. He also states, "Lastly, just a side note, but you can't get away from GMO's. As of 2012, 88% of corn and 94% of soy crops in the US were genetically modified..." James gives this statistic to show that many people eat GMO's constantly without realizing, even though they claim it is harmful.

In conclusion, the short term effects of genetically modified foods show that they can cause rare diseases, and public surveys shows that people are very weary and unsure of the effects of genetically modified foods, while other studies have shown that there are no harmful differences between GMO's and non-GMO's, and are shown to have pros including: more nutritious food, tastier food, increased food supply with a longer shelf life, and less use of pesticides. Future research will evolve into looking more into the long-term effects of genetically modified foods, and how they affect the health of the general public.